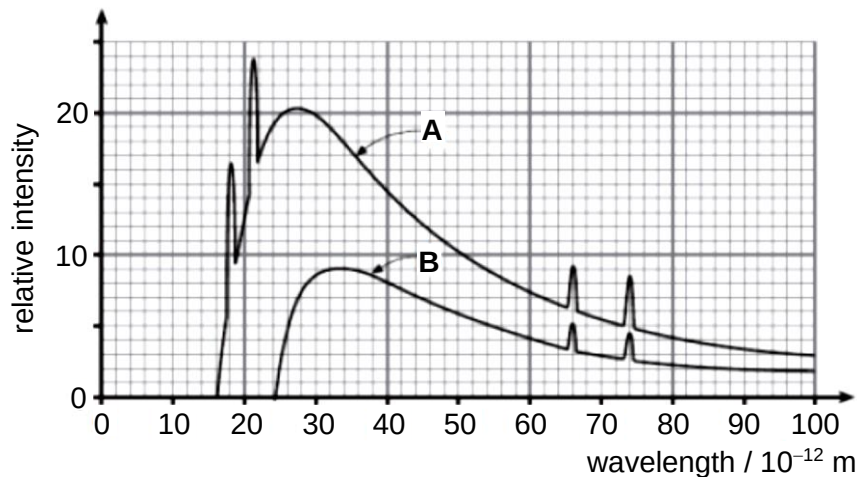


28 The diagram shows two spectra of X-rays from an X-ray tube.



Which of the following statements is true?

- A** The accelerating potential to produce spectrum B is lower than that to produce spectrum A.
- B** A different target metal is used to produce spectra A and B as shown from the existence of additional peaks in spectrum A.

- C** With the same accelerating potential, the temperature of cathode used to produce spectrum A is higher than that to produce spectrum B as shown from the higher intensity of X-ray photons in spectrum A.
- D** The temperature of cathode used to produce spectrum A is higher than that to produce spectrum B as shown from the existence of additional peaks in spectrum A.

Ans: A

A: Lower accelerating p.d. for B, smaller loss in EPE and smaller gain in KE, smaller E of most energetic X-ray photon produced, larger cut-off wavelength. At the same time, fewer X-ray photons produced, hence intensity is lowered.

B: Additional peaks are observed in A because the cut-off wavelength for A is shorter. Same target metal used.

C: Higher temperature of cathode leads to more X-ray photons produced, hence intensity is higher. But it does not explain why the cut-off wavelength for A is shorter.

D: Higher temperature of cathode leads to more X-ray photons produced, hence intensity is higher. But it does not explain why additional peaks for A are observed.